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Connect a GitHub Repo with AWS



sahini sourya

```
Complete!  
[ec2-user@ip-172-31-9-131 ~]$ git --version  
git version 2.50.1  
[ec2-user@ip-172-31-9-131 ~]$ █
```



Introducing Today's Project!

In this project, I will demonstrate Github repository set up for web application code. This project is my day 2 of 7 days Devops learning challenge. I'm doing this project to learn git with github that will help me to build web application through Github repositories

Key tools and concepts

Services I used were Github, amazon EC2 which was my development instance, keypairs, VS code. Key concepts I learnt include setting up Github repository, difference between Git and Github, the commands for staging, saving and pushing. I also learnt about upstream repositories and setting up an remote origin and branches.

Project reflection

This project took me approximately 2 hours, The most challenging part was to learn all the Git commands and their heirarchy of usages, how important braching and what commands to use it. It was most rewarding to setting up readme file that will useful for Github repository explanation who would visit and understand very well.

placeholder

I did this project because to learn more about Git and Github and its day 2 of 7 day DevOps challenge. I am excited now as i have repository that stores my web app source code.

This project is part two of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project tommorow.



Git and GitHub

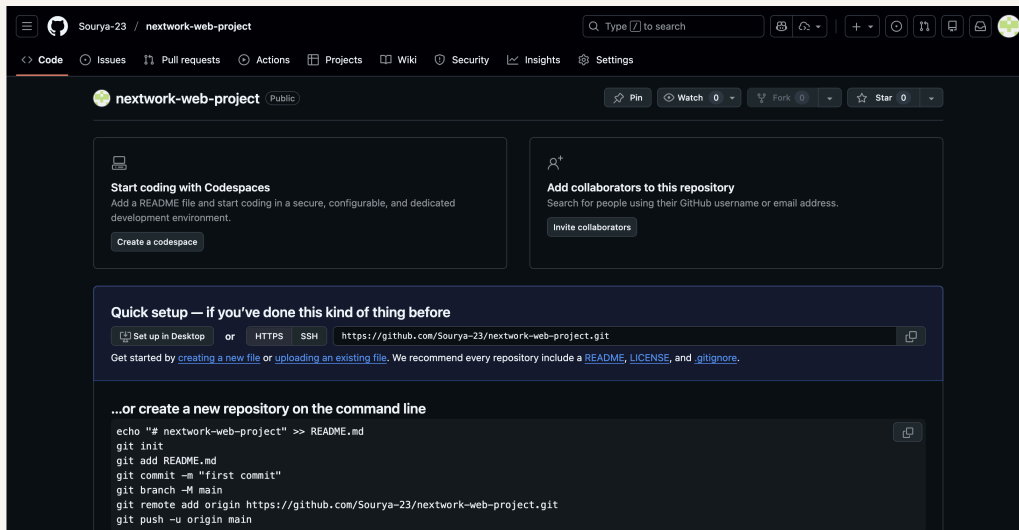
Git is version control system that tracks every change that has been made, and lets developer go back to an earlier version of work if something break. it's also inccredibly helpful in collaboration, with git we can see who make what change and handover from development to deployment is lot smoother with everyone looking at same shared repository of code. I installed Git using the command "sudo dnf install git -y".

GitHub is a platform to collaborate that lets us store and share on our code.It's called Github cause it uses git as tool for version control. Github adds much more that collaboration, seeing public repositories and managing multiple repositories/projects makes much easier. I'm using GitHub in this project to store our web application code.



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My local repository

A Git repository is an online folder that you can use to store web application code and all versions of the code. Think of it as a place for all the code for updates and changes of the specific project.

Git init is a command that initializes git in my local repository. I ran git init in my "web app project" folder, which tells terminal that I want to start tracking changes locally.

A branch in Git is parallel versions of the same project. For example, if you wanted to test a change to your code, you can set up a new branch that lets you diverge from the original/main version of your code (called master) so you can experiment with new features or test bug fixes safely. After running git init, the response from the terminal was we initialized Git, and by default we are using the 'main' branch.



```
[ec2-user@ip-172-31-9-131 nextwork-web-project]$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/ec2-user/nextwork-web-project/.git/
```



To push local changes to GitHub, I ran three commands

git add

The first command I ran was "git add.", which adds my changes to staging area. A staging area is like a place to review all changes made to the code, so I can decide what changes I like to save.

git commit

The second command I ran was "git commit" which is a command to save saving changes in the staging area. Using '-m' means also leaving message for the commit. for example "made changes to index.jsp".

git push

The third command I ran was "git push -u origin master" which means pushing code changes I made to remote origin. Using '-u' means is a specific flag that makes our remote origin default upstream repository. Next time if I have any changes, I simply run "git push" and git already knows where to push those changes.



Authentication

When I commit changes to GitHub, Git asks for my credentials because it needs to authenticate me before letting me code change to the repository, it needs to do this to know I have the right to change the code.

Local Git identity

Git needs my name and email because it's a version control system which means tracking changes to the code and really identifies who made what change to piece of code. To really identify who that made changes, Git will use your name and email.

Running git log showed me that by default, Git is saving our details for code changes to a username called "EC2 Default User" instead of actual name or details.

```
[ec2-user@ip-172-31-9-131 nextwork-web-project]$ git log
commit 954c1906ce0cb31b8242bd39247856d84219c4b3 (HEAD -> master, origin/master)
Author: EC2 Default User <ec2-user@ip-172-31-9-131.ap-south-2.compute.internal>
Date: Mon Jan 5 10:37:08 2026 +0000

    Updated index.jsp with new content
[ec2-user@ip-172-31-9-131 nextwork-web-project]$
```



GitHub tokens

GitHub authentication failed when I entered my password because password authentication was already removed in 2021 because there are too many security risks associated with my password over the terminal to Github. So more secure ways are available

A GitHub token is like a temporary password that wants access to my Github account. I'm using one in this project because it leads me safely to authenticate to Github repository while in my EC2 instance,

I could set up a GitHub token by visiting the developer setting in Github and I set one up that expires in 30 days and only allows permissions to repositories



GitHub Apps

OAuth Apps

Personal access tokens

Fine-grained tokens

Tokens (classic)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

Generated for EC2 Instance Access. This is a part of NextWork's

What's this token for?

Expiration

30 days (Feb 04, 2026)

The token will expire on the selected date

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes.](#)

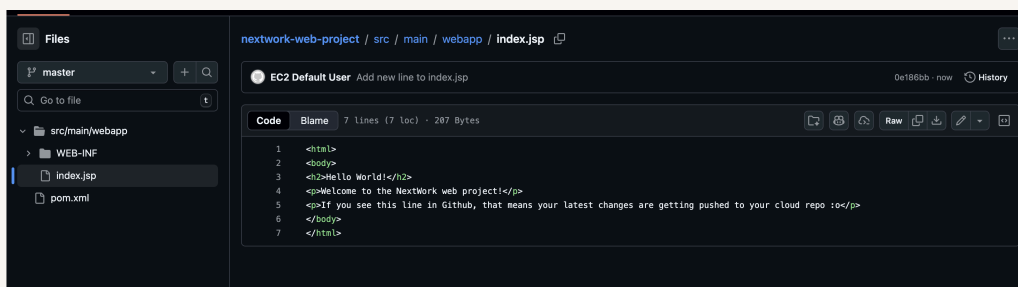
☒ repo	Full control of private repositories
☒ repo:status	Access commit status
☒ repo_deployment	Access deployment status
☒ public_repo	Access public repositories
☒ repo:invite	Access repository invitations
☒ security_events	Read and write security events



Making changes again

I wanted to see Git working in action, so I made another change in `index.jsp`. I couldn't see the changes in my GitHub repo initially because I hadn't committed or push my changes to Github.

I finally saw the changes in my GitHub repo after running same three commands "git add. ", "git commit", "git staged" and then refreshing `index.jsp` in our repository to confirm the changes on Github repository.



The screenshot shows a code editor interface with a dark theme. On the left, a file explorer shows the project structure: `src/main/webapp` containing `WEB-INF` (with `index.jsp` inside) and `pom.xml`. The main editor area displays the content of `index.jsp` for the `EC2 Default User` branch. The file path is `nextwork-web-project / src / main / webapp / index.jsp`. The commit message is "Add new line to index.jsp" with hash `0e186bb`. The code content is as follows:

```
1 <html>
2 <body>
3 <h1>Hello World!</h1>
4 <p>Welcome to the NextWork web project!</p>
5 <p>If you see this line in Github, that means your latest changes are getting pushed to your cloud repo :o</p>
6 </body>
7 </html>
```

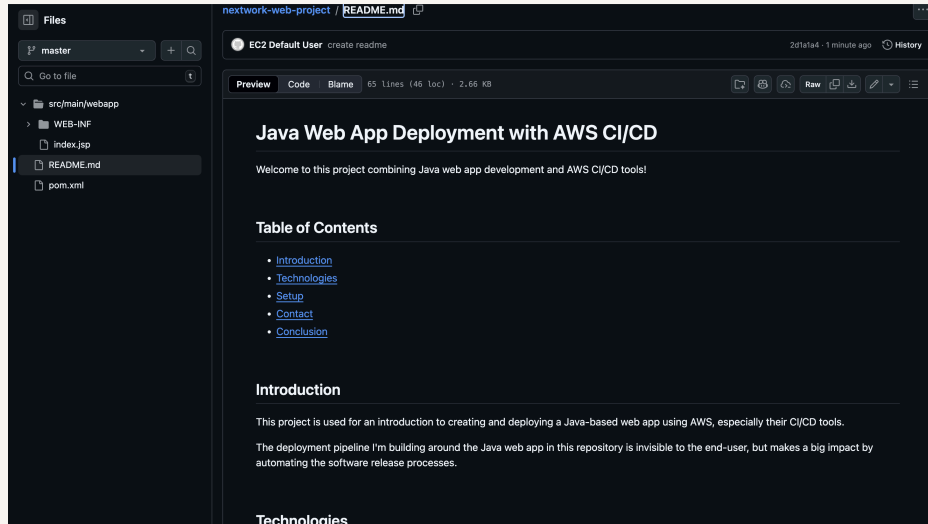


Setting up a README file

As a finishing touch to my GitHub repository, I added a README file, which is a document that introduces and explains our project, like what the project does, how to set it up, and how to use it. I added a README file by using "touch README.md" command in my terminal with my nextwork-webproject directory telling terminal create a Markdown file named README.

My README is written in Markdown because readme file contains the format like headings, lists, images, links and even code lines to my documentation without needing complex code. with explanation text Special characters can help you format text in Markdown, such as '#' for headings, '**text**' to look bold, [link text](URL) to create a hyperlink, - or * for unordered bullet lists 1., 2., etc., for ordered lists, ![image captions](image_URL) to embed an image, Put text inbetween `backticks` for inline code like this, Use triple backticks ``` for multi-line code blocks, etc.,

My README file has 6 sections that outline table of contents, an Introduction ,Technologies(i.e, AWS services that i used),how to set it up , how to set up repository locally, and then a section for contacting myself and thank you message at end.





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